Separating DNA Fragments

Biology 2e chapter 17 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

DNA fragments move through an agarose gel in an electric field. Compare unknown bands with a size ladder on the same gel. Within a useful range, migration distance relates approximately to the logarithm of fragment size.

Investigation question

How can migration distance help estimate DNA fragment size and compare samples?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

Run the gel, switch power off, and stain before measuring. Lanes are ladder, reference A, reference B and unknown. Use the on-screen ruler and ladder to estimate the three unknown bands; matching bands alone do not identify a person.

Prepare the equipment in the displayed order. Observe or classify the sample, review your measurements, then click Collect data and continue. Repeat through the assigned comparison. Explore other settings after collecting the full comparison.

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Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: base pairs

Condition / sample	Repeat	Source	Largest band (bp)	Middle band (bp)	Smallest band (bp)
Gel run 1	1				
Gel run 2	1				

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Sample / field: _____ Source: _____

Item / band / cell	Measurement or classification	Reasoning / uncertainty

For large counts, use tallies with a clear starting position so each specimen is counted once. For microscopy and gels, record calibration and measurement units.

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Estimate unknown sizes from the ladder and compare the reference patterns.

Why use a ladder in each gel rather than a fixed distance-to-size conversion?

What limits the conclusion from a matching pattern?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Original band patterns and a simplified log-distance model; no supplier mystery or real forensic sample. Diffusion, DNA topology, overloading and nonlinear extremes are omitted. Illustrative voltage and timing are not a physical equipment protocol.